

7 Separation of Variables

Chapter 5, An Introduction to Partial Differential Equations, Pichover and Rubinstein

In this section we introduce the technique, called the method of separations of variables, for solving initial boundary value-problems.

7.1 Heat Equation

We consider the heat equation satisfying the initial conditions

$$\begin{cases} u_t = ku_{xx}, & x \in [0, L], t > 0 \\ u(x, 0) = \phi(x), & x \in [0, L] \end{cases} \quad (7.1)$$

We seek a solution u satisfying certain boundary conditions. The boundary conditions could be as follows:

- (a) *Dirichlet* $u(0, t) = u(L, t) = 0$.
- (b) *Neumann* $u_x(0, t) = u_x(L, t)$.
- (c) *Periodic* $u(-L, t) = u(L, t)$ and $u_x(-L, t) = u_x(L, t)$.

We look for solutions of the form

$$u(x, t) = X(x)T(t)$$

where X and T are function which have to be determined. Substituting $u(x, t) = X(x)T(t)$ into the equation, we obtain

$$X(x)T'(t) = kX''(x)T(t)$$

from which, after dividing by $kX(x)T(t)$, we get

$$\frac{T'(t)}{kT(t)} = \frac{X''(x)}{X(x)}.$$

The left side depends only on t whereas the right hand side depends only on x . Since they are equal, they must be equal to some constant $-\lambda$. Thus

$$T' + \lambda kT = 0 \quad (7.2)$$

and

$$X'' + \lambda X = 0. \quad (7.3)$$

The general solution of the first equation is given

$$T(t) = Be^{-\lambda kt}$$

for an arbitrary constant B . The general solutions of the second equation are as follows.

- (1) If $\lambda < 0$, then $X(x) = \alpha \cosh \sqrt{-\lambda}x + \beta \sinh \sqrt{-\lambda}x$.
- (2) If $\lambda = 0$, then $X(x) = \alpha x + \beta$.
- (3) If $\lambda > 0$, then $X(x) = \alpha \cos \sqrt{\lambda}x + \beta \sin \sqrt{\lambda}x$.

(7.4)

In addition, the function X which solves the second equation will satisfy boundary conditions depending on the boundary condition imposed on u . The problem

$$\begin{cases} X'' + \lambda X = 0 \\ X \text{ satisfies boundary conditions} \end{cases} \quad (7.5)$$

is called the *eigenvalue problem*, a nontrivial solution is called an *eigenfunction* associated with the *eigenvalue* λ .

7.1.1 Heat equation with Dirichlet boundary conditions

We consider (7.1) with the Dirichlet condition

$$u(0, t) = u(L, t) = 0 \quad \text{for all } t \geq 0.$$

In this case the eigenvalue problem (7.5) becomes

$$\begin{cases} X'' + \lambda X = 0 \\ X(0) = X(L) = 0. \end{cases} \quad (7.6)$$

We have to find nontrivial solutions X of the eigenvalue problem (7.6). If $\lambda = 0$, then $X(x) = \alpha x + \beta$ and $0 = X(0) = \alpha \cdot 0 + \beta$ implies that $\beta = 0$ and $0 = X(L) = \alpha L$ implies that $\alpha = 0$. If $\lambda < 0$. Then $0 = X(0) = \alpha \cosh 0 + \beta \sinh 0 = \alpha$ and $0 = X(L) = \beta \sinh L$ shows that also $\beta = 0$. We conclude that $\lambda \leq 0$ is not an eigenvalue of the problem (7.5). Finally, consider $\lambda > 0$. Then $0 = X(0) = \alpha \cos \sqrt{\lambda} \cdot 0 = \alpha$ and $0 = X(L) = \beta \sin \sqrt{\lambda}L$. Since X is nontrivial solution, $\beta \neq 0$ and hence $\sin \sqrt{\lambda}L = 0$. Consequently,

$$\lambda = \left(\frac{n\pi}{L}\right)^2, \quad n \geq 1$$

and the corresponding eigenfunction is given by

$$X_n(x) = \sin \frac{n\pi x}{L}$$

After substituting $\lambda = (n\pi/L)^2$ to (7.2), we get the family of solutions

$$T_n(t) = B_n e^{-k(\frac{n\pi}{L})^2 t}.$$

Thus we have obtained the following sequence of solutions

$$u_n(x, t) = X_n(x)T_n(t) = B_n \sin \frac{n\pi x}{L} e^{-k(\frac{n\pi}{L})^2 t}.$$

We obtain more solutions by taking linear combinations of the u_n 's (recall the superposition principle)

$$u(x, t) = \sum_{n=1}^N u_n(x, t) = \sum_{n=1}^N B_n \sin \frac{n\pi x}{L} e^{-k(\frac{n\pi}{L})^2 t},$$

and then by passing to the limit $N \rightarrow \infty$,

$$u(x, t) = \sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{L} e^{-k(\frac{n\pi}{L})^2 t}. \quad (7.7)$$

Finally, we consider the initial condition. At $t = 0$, we must have

$$u(x, 0) = \sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{L} = \phi(x). \quad (7.8)$$

The coefficients, B_n can be computed as follows. Fix $m \in \mathbb{N}$. Multiplying the above equality by $\sin \frac{m\pi x}{L}$ and then integrating over $[0, L]$, we get

$$\begin{aligned} \int_0^L \phi(x) \sin \frac{m\pi x}{L} dx &= \int_0^L \sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{L} \sin \frac{m\pi x}{L} dx \\ &= \sum_{n=1}^{\infty} B_n \int_0^L \sin \frac{n\pi x}{L} \sin \frac{m\pi x}{L} dx. \end{aligned}$$

(above we ignore the question about interchanging $\sum_{n=1}^{\infty}$ with $\int_0^L \cdot$.) Since

$$\int_0^L \sin \frac{n\pi x}{L} \sin \frac{m\pi x}{L} dx = \begin{cases} 0 & n \neq m \\ \frac{L}{2} & n = m, \end{cases} \quad (7.9)$$

we find that

$$B_m = \frac{2}{L} \int_0^L \phi(x) \sin \frac{m\pi x}{L} dx. \quad (7.10)$$

Example 7.1. Consider the problem,

$$\begin{aligned} u_t - u_{xx} &= 0, \quad 0 < x < \pi, \quad t > 0 \\ u(0, t) &= u(\pi, t) = 0, \quad t \geq 0 \\ u(x, 0) &= \phi(x) = \begin{cases} x & 0 \leq x \leq \pi/2 \\ \pi - x & \pi/2 \leq x \leq \pi. \end{cases} \end{aligned} \quad (7.11)$$

Here $k = 1$ and $[0, L] = [0, \pi]$. In view of (7.8), the solution $u(x, t)$ is given by

$$u(x, t) = \sum_{n=1}^{\infty} B_n (\sin nx) e^{-n^2 t}.$$

Using (7.10), the coefficients B_n are equal to

$$\begin{aligned} B_n &= \frac{2}{\pi} \int_0^{\pi} \phi(x) \sin nx \, dx \\ &= \frac{2}{\pi} \int_0^{\pi/2} x \sin nx \, dx + \frac{2}{\pi} \int_{\pi/2}^{\pi} (\pi - x) \sin nx \, dx. \end{aligned}$$

Integrating by parts we find that the right-hand side is equal to

$$\frac{2}{\pi} \left[\frac{-x \cos nx}{n} + \frac{\sin x}{n^2} \right]_0^{\pi/2} + \frac{2}{\pi} \left[\frac{-(\pi - x) \cos nx}{n} - \frac{\sin x}{n^2} \right]_{\pi/2}^{\pi} = \frac{4}{\pi n^2} \sin \frac{n\pi}{2}.$$

Since

$$\sin \frac{n\pi}{2} = \begin{cases} 0 & n = 2k \\ (-1)^{k+1} & n = 2k - 1 \end{cases}$$

for $k \geq 1$, we get

$$\begin{aligned} u(x, t) &= \sum_{n=1}^{\infty} B_n (\sin nx) e^{-n^2 t} \\ &= \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{(2n-1)^2} \sin[(2n-1)x] e^{-(2n-1)^2 t}. \end{aligned} \quad (7.12)$$

We claim that u is indeed the classical solution of the equation (7.11). To see this, denote by

$$u_n(x, t) = \frac{(-1)^{n+1}}{(2n-1)^2} \sin[(2n-1)x] e^{-(2n-1)^2 t}$$

so that $u(x, t) = \frac{4}{\pi} \sum_n u_n(x, t)$. Since

$$|u_n(x, t)| = \left| \frac{(-1)^{n+1}}{(2n-1)^2} \sin[(2n-1)x] e^{-(2n-1)^2 t} \right| \leq \frac{1}{(2n-1)^2}$$

and the series $\sum_n \frac{1}{(2n-1)^2}$ converges. Hence, in view of the Weierstrass M-test (see Review), the series on the right hand side converges uniformly to a continuous function u in the region $D = \{(x, t) | 0 \leq x \leq \pi, t \geq 0\}$. It remain to show that u is differentiable with respect to t and twice differentiable with respect to x , and satisfies the heat equation. Take $\varepsilon > 0$, and let

$$D_\varepsilon = \{(x, t) | 0 < x < \pi, t > \varepsilon\}.$$

Differentiating u_n with respect to t we get

$$|u_n(x, t)_t| = \left| \frac{(2n-1)^2}{(2n-1)^2} \sin[(2n-1)x] e^{-(2n-1)^2 t} \right| \leq e^{-(2n-1)^2 \varepsilon}.$$

The series $\sum_n e^{-(2n-1)^2 \varepsilon}$ converges (apply the ratio test). Hence applying the Weierstrass M-test again we find that the series of $u_n(x, t)_t$ converges uniformly to a continuous function on D_ε . Similarly, differentiating with respect to x twice we get

$$|u_n(x, t)_{xx}| = \left| \frac{(2n-1)^2}{(2n-1)^2} \sin[(2n-1)x] e^{-(2n-1)^2 t} \right| \leq e^{-(2n-1)^2 \varepsilon}$$

and again $\sum_n u_n(x, t)_{xx}$ converges uniformly on D_ε . Hence, using that $(u_n)_t - (u_n)_{xx} = 0$,

$$u_t - u_{xx} = \sum_n (u_n)_t - \sum_n (u_n)_{xx} = \sum_n [(u_n)_t - (u_n)_{xx}] = 0.$$

So, u is a solution on D_ε and since ε was arbitrary, u is a classical solution on D .

7.1.2 Heat equation with Neumann boundary conditions

We consider the heat equation (7.1) but with Neumann boundary conditions

$$u_x(0, t) = u_x(\pi, t) = 0 \quad \text{for all } t \geq 0.$$

In this case the eigenvalue problem (7.5) becomes

$$\begin{cases} X'' + \lambda X = 0 \\ X'(0) = X'(\pi) = 0 \end{cases} \quad (7.13)$$

As before the problem doesn't have negative eigenvalues. If $\lambda = 0$, the general solution is $X(x) = \alpha x + \beta$ so that $0 = X'(0) = \beta$, implies that $\lambda_0 = 0$ is an eigenvalue with the unique (up to multiplication by a constant) eigenfunction $X_0(x) \equiv 1$. If $\lambda > 0$, then the general solution of the problem is $X(x) = \alpha \cos \sqrt{\lambda}x + \beta \sin \sqrt{\lambda}x$ form which we conclude that $0 = X'(0) = \beta$ and $0 = X'(L) = -\sqrt{\lambda}\alpha \sin \sqrt{\lambda}L$ implies that $\lambda > 0$ is an eigenvalue if and only if

$$\lambda = \lambda_n = \left(\frac{n\pi}{L}\right)^2, \quad n \geq 1$$

and the corresponding eigenfunction $X_n(x)$ is given by

$$X_n(x) = \cos \frac{n\pi x}{L}.$$

Then the corresponding solutions of $T' + \lambda k T = 0$ are

$$\begin{aligned} T_0(t) &= B_0 \\ T_n(t) &= B_n e^{-k\left(\frac{n\pi}{L}\right)^2 t}, \quad n \geq 1. \end{aligned}$$

Thus we obtain a sequence of solutions

$$u_n(x, t) = B_n e^{-k\left(\frac{n\pi}{L}\right)^2 t} \cos \frac{n\pi x}{L}, \quad n \geq 0$$

which we combine to form the series

$$u(x, t) = \sum_{n \geq 0} B_n e^{-k\left(\frac{n\pi}{L}\right)^2 t} \cos \frac{n\pi x}{L}.$$

At $t = 0$, we have

$$\phi(x) = u(x, 0) = \sum_{n \geq 0} B_n \cos \frac{n\pi x}{L}.$$

To compute B_m 's, multiply this equality by $\cos \frac{m\pi x}{L}$ and integrate over $[0, L]$. Then

$$\int_0^L \phi(x) \cos \frac{m\pi x}{L} dx = \sum_{n \geq 0} B_n \int_0^L \cos \frac{n\pi x}{L} \cos \frac{m\pi x}{L} dx.$$

Since

$$\int_0^L \cos \frac{m\pi x}{L} dx = \begin{cases} L & m = 0 \\ 0 & m \geq 1, \end{cases} \quad (7.14)$$

and

$$\int_0^L \cos \frac{n\pi x}{L} \cos \frac{m\pi x}{L} dx = \begin{cases} 0 & n \neq m \\ \frac{L}{2} & n = m, \end{cases} \quad (7.15)$$

it follows that

$$B_0 = \frac{1}{L} \int_0^L \phi(x) dx \quad \text{and} \quad B_m = \frac{2}{L} \int_0^L \phi(x) \cos \frac{m\pi x}{L} dx, \quad m \geq 1.$$

7.1.3 Heat equation with periodic boundary conditions

Next we consider (7.1) with the periodic boundary conditions

$$u(-L, t) = u(L, t) \quad \text{and} \quad u_x(-L, t) = u_x(L, t) \quad \text{for all } t \geq 0.$$

In this case the eigenvalue problem takes the form

$$\begin{cases} X'' + \lambda X = 0 \\ X(0) = X(L), X'(0) = X'(L) \end{cases} \quad (7.16)$$

This follows from $X(-L)T(t) = X(L)T(t)$ and $X'(-L)T(t) = X'(L)T(t)$. To find eigenvalues, we first consider $\lambda < 0$. From (7.4) and since $\sinh$ is odd and $\cosh$ is even, the condition $X(-L) = X(L)$ implies that

$$\beta \sinh \sqrt{-\lambda} L = 0$$

so that $\beta = 0$. The condition $X'(-L) = X'(L)$ implies that

$$\alpha \sinh \sqrt{-\lambda} L = 0$$

so that $\alpha = 0$. If $\lambda = 0$, then $X(-L) = -\alpha L + \beta = \alpha L + \beta = X(L)$ so that $\alpha = 0$. So $\lambda_0 = 0$ is an eigenvalue with the corresponding eigenfunction $X_0(x) \equiv 1$. Finally, let $\lambda > 0$. Then $X(-L) = X(L)$ gives either $\beta = 0$ or $\sqrt{\lambda} = \frac{n\pi}{L}$ and the condition $X'(-L) = X'(L)$ gives either $\alpha = 0$ or $\sqrt{\lambda} = \frac{n\pi}{L}$. Hence the positive eigenvalues are

$$\lambda_n = \left(\frac{n\pi}{L} \right)^2, \quad n \geq 1$$

and the corresponding eigenfunctions are

$$X_n(x) = B_n \cos \left(\frac{n\pi x}{L} \right) + C_n \sin \left(\frac{n\pi x}{L} \right).$$

Thus the product solutions of the periodic boundary problem are

$$\begin{aligned} u_0(x, t) &= A_0 \\ u_n(x, t) &= \left(B_n \cos \left(\frac{n\pi x}{L} \right) + C_n \sin \left(\frac{n\pi x}{L} \right) \right) e^{-\left(\frac{n\pi}{L} \right)^2 t} \end{aligned}$$

which can be combined to form the series

$$u(x, t) = A_0 + \sum_{n \geq 1} \left(B_n \cos \left(\frac{n\pi x}{L} \right) + C_n \sin \left(\frac{n\pi x}{L} \right) \right) e^{-\left(\frac{n\pi}{L} \right)^2 t}.$$

At $t = 0$, we have

$$\phi(x) = u(x, 0) = A_0 + \sum_{n \geq 1} \left(B_n \cos \left(\frac{n\pi x}{L} \right) + C_n \sin \left(\frac{n\pi x}{L} \right) \right).$$

Integrating over $[-L, L]$, we get

$$A_0 = \frac{1}{2L} \int_{-L}^L \phi(x) dx$$

since $\int_{-L}^L \cos \left(\frac{n\pi x}{L} \right) dx = \int_{-L}^L \sin \left(\frac{n\pi x}{L} \right) dx = 0$. Next multiplying both sides by $\cos \left(\frac{m\pi x}{L} \right)$ and integrating over $[-L, L]$ leads to

$$B_m = \frac{1}{L} \int_{-L}^L \phi(x) \cos \left(\frac{m\pi x}{L} \right) dx$$

since

$$\begin{aligned} \int_{-L}^L \cos \left(\frac{n\pi x}{L} \right) \cos \left(\frac{m\pi x}{L} \right) dx &= \begin{cases} 0 & n \neq m \\ L & n = m \end{cases} \\ \int_{-L}^L \sin \left(\frac{n\pi x}{L} \right) \cos \left(\frac{m\pi x}{L} \right) dx &= 0, \quad \text{all } n, m \geq 1. \end{aligned}$$

Finally, multiplying both sides by $\sin \left(\frac{m\pi x}{L} \right)$ and integrating over $[-L, L]$ leads to

$$C_m = \frac{1}{L} \int_{-L}^L \phi(x) \sin \left(\frac{m\pi x}{L} \right) dx$$

since

$$\int_{-L}^L \sin \left(\frac{n\pi x}{L} \right) \sin \left(\frac{m\pi x}{L} \right) dx = \begin{cases} 0 & n \neq m \\ L & n = m. \end{cases}$$

7.2 Separation of variable for the wave equation

The above techniques can be used to solve the wave equation. We focus on the wave equation satisfying Dirichlet boundary condition (Neumann boundary conditions are treated in the book). We consider the initial boundary problem

$$\begin{aligned} u_{tt} - c^2 u_{xx} &= 0, & 0 < x < L, t > 0 \\ u(0, t) = u(L, t) &= 0, & t \geq 0 \\ u(x, 0) &= \phi, & 0 < x < L \\ u_t(x, 0) &= \psi, & 0 < x < L. \end{aligned} \tag{7.17}$$

If we take $u(x, t) = X(x)T(t)$, then (7.17) becomes

$$X(x)T''(t) = c^2 X''(x)T(t).$$

Dividing by $c^2 X(x)T(t)$, we get

$$\frac{T''(t)}{c^2 T(t)} = \frac{X''(x)}{X(x)}$$

from which we conclude that both sides of this equality must be equal to a constant $-\lambda$. Thus, we obtain two second order differential equations

$$X'' + \lambda X = 0 \tag{7.18}$$

$$T'' + \lambda c^2 T = 0. \tag{7.19}$$

The boundary conditions imply that

$$X(0) = X(L) = 0.$$

so that the function X should be a solution of the eigenvalue problem,

$$\begin{aligned} X'' + \lambda X &= 0, & 0 < x < L \\ X(0) = X(L) &= 0. \end{aligned}$$

Just as in the case of the heat equation the eigenvalues are and the associated eigenfunctions are

$$\lambda_n = \left(\frac{n\pi}{L}\right)^2, \quad X_n(x) = \sin \frac{n\pi x}{L} \quad n \geq 1.$$

Then the solution of (7.19) with $\lambda = \lambda_n$ is of the form

$$T_n(t) = A_n \cos\left(\frac{n\pi ct}{L}\right) + B_n \sin\left(\frac{n\pi ct}{L}\right)$$

where A_n and B_n are constants. The product solutions of the boundary value problem are given by

$$u_n(x, t) = \left(A_n \cos \left(\frac{n\pi ct}{L} \right) + B_n \sin \left(\frac{n\pi ct}{L} \right) \right) \sin \frac{n\pi x}{L}$$

which can be combined in the series

$$u(x, t) = \sum_{n \geq 1} \left(A_n \cos \left(\frac{n\pi ct}{L} \right) + B_n \sin \left(\frac{n\pi ct}{L} \right) \right) \sin \frac{n\pi x}{L}. \quad (7.20)$$

Setting $t = 0$, we get

$$\phi(x) = u(x, 0) = \sum_{n \geq 1} A_n \cos \left(\frac{n\pi ct}{L} \right) \sin \frac{n\pi x}{L}.$$

Multiplying by $\sin \frac{m\pi x}{L}$ and integrating over $[0, L]$ we find that

$$A_n = \frac{2}{L} \int_0^L \phi(x) \sin \frac{n\pi x}{L} dx.$$

Next, differentiate (7.20) with respect to t at $t = 0$ to get

$$\psi(x) = u_t(x, 0) = \frac{n\pi c}{L} \sum_{n \geq 1} B_n \sin \frac{n\pi x}{L}.$$

To compute B_n 's, multiply both sides by $\sin \frac{m\pi x}{L}$ and integrate over $[0, L]$ to get

$$B_m = \frac{2}{L} \frac{L}{m\pi c} \int_0^L \phi(x) \sin \frac{m\pi x}{L} dx = \frac{2}{m\pi c} \int_0^L \phi(x) \sin \frac{m\pi x}{L} dx.$$

Example 7.2. Consider

$$\begin{cases} u_{tt} - 4u_{xx} = 0 & (x, t) \in (0, \pi) \times (0, \infty) \\ u(0, t) = u(\pi, t) = 0 & t \geq 0 \\ u(x, 0) = \sin 2x, \quad u_t(x, 0) = 0, & 0 \leq x \leq \pi \end{cases}$$

Here $c = 2$, $[0, L] = [0, \pi]$. The formal solution is of the form

$$u(x, t) = \sum_{n \geq 1} (A_n \cos 2nt + B_n \sin 2nt) \cdot \sin nx.$$

From the above formulae, $B_n = 0$ since $\psi \equiv 0$ and

$$A_n = \frac{2}{L} \int_0^L \sin 2x \sin nx dx = \begin{cases} 1 & n = 2 \\ 0 & \text{otherwise.} \end{cases}$$

Consequently,

$$u(x, t) = \cos 4t \sin 2x.$$

7.3 The energy method and uniqueness

In this section we use the so called energy method to prove the uniqueness of initial boundary value problems. The method is based on the physical principle of energy conservation.

Theorem 7.3. *Consider the Dirichlet problem*

$$\begin{cases} u_t - ku_{xx} = f(x, t) & x \in [0, L], t > 0 \\ u(0, t) = a(t), u(L, t) = b(t) & t \geq 0 \\ u(x, 0) = \phi(x) & x \in [0, L]. \end{cases}$$

Then the problem has at most one solution.

Proof. Suppose that the problem has two solutions u_1 and u_2 . Setting $v = u_1 - u_2$, the function v solves

$$\begin{cases} v_t - kv_{xx} = 0 & x \in [0, L], t > 0 \\ v(0, t) = 0, v(L, t) = 0 & t \geq 0 \\ v(x, 0) = 0 & x \in [0, L]. \end{cases}$$

Define the energy

$$E(t) = \frac{1}{2} \int_0^L v(x, t)^2 dx.$$

Differentiating with respect to t , we get

$$E'(t) = \int_0^L v_t(x, t)v(x, t) dx = \int_0^L v(x, t)v_{xx}(x, t) dx.$$

Integrating by part and using the boundary conditions, we find that

$$E'(t) \int_0^L v(x, t)v_{xx}(x, t) dx = [kv(x, t)v_x(x, t)]_0^L - \int_0^L kv_x(x, t)^2 dx \leq 0.$$

Hence the energy E is decreasing. Since $E(0) = 0$ and $E(t) \geq 0$, we get that $E \equiv 0$. This implies that $v(\cdot, t) \equiv 0$ for every t . Consequently, $u_1(x, t) = u_2(x, t)$ for all $x \in [0, L]$ and $t \geq 0$. ■

Theorem 7.4. *Consider the Neumann problem*

$$\begin{cases} u_{tt} - c^2u_{xx} = f(x, t) & x \in [0, L], t > 0 \\ u_x(0, t) = a(t), u_x(L, t) = b(t) & t \geq 0 \\ u(x, 0) = \phi(x) & x \in [0, L]. \\ u_t(x, 0) = \psi(x) & x \in [0, L]. \end{cases}$$

Then the problem has at most one solution.

Proof. Suppose that the problem has two solutions u_1 and u_2 . Setting $v = u_1 - u_2$, the function v solves

$$\begin{cases} v_{tt} - c^2 v_{xx} = 0 & x \in [0, L], t > 0 \\ v_x(0, t) = 0, v_x(L, t) = 0 & t \geq 0 \\ v(x, 0) = 0 & x \in [0, L]. \\ v_t(x, 0) = 0 & x \in [0, L]. \end{cases}$$

Define the energy

$$E(t) = \frac{1}{2} \int_0^L (v_t(x, t)^2 + c^2 v_x(x, t)^2) dx.$$

Differentiating with respect to t , we get

$$E'(t) = \int_0^L (v_t(x, t)v_{tt}(x, t) + c^2 v_x(x, t)v_{xt}(x, t)) dx. \quad (7.21)$$

Note that

$$c^2 v_x v_{xt} = c^2 [(v_x v_t)_x - v_{xx} v_t] = c^2 (v_x v_t)_x - v_{tt} v_t$$

where we have used $v_{tt} - c^2 v_{xx} = 0$. Substituting this into (7.21) and using the boundary conditions, we get

$$E'(t) = c^2 \int_0^L \frac{\partial}{\partial x} (v_x(x, t)v_t(x, t)) dx = c^2 [v_x(x, t)v_t(x, t)]_0^L = 0.$$

Hence $E(t) = \text{constant}$ for all t . At $t = 0$, we have $v(x, 0) = 0$ so that $v_x(x, 0) = 0$. Since $v_t(x, 0) = 0$, we conclude that $E(0) = 0$ and so $E(t) = 0$ for all $t \geq 0$. This implies that

$$v_t(x, t)^2 + c^2 v_x(x, t)^2 = 0$$

for all $x \in [0, T]$ and $t \geq 0$. Thus, $v_t(x, t) = v_x(x, t) = 0$ for all $x \in [0, T]$ and $t \geq 0$. Consequently, $v \equiv \text{constant}$. But then $v(x, 0) = 0$, implies that $v \equiv 0$. ■

Review–Infinite series of functions

Formally, an infinite series $\sum_{n \geq 1} a_n$ is a pair consisting of a sequence (a_n) and a sequence (s_n) of partial sums, $s_n = \sum_{k=1}^n a_k$. The series $\sum a_n$ converges, provided that a sequence (s_n) converges, that is, there is s such that

$\lim s_n = s$. The number s is called the value of the series $\sum a_n$ and is denoted by $s = \sum_{n=1}^{\infty} a_n$. The series $\sum a_n$ is said to converge absolutely, if $\sum |a_n|$ converges. If $\sum |a_n|$ converges, then $\sum a_n$ converges. However, the convergence of $\sum a_n$ does not guarantee convergence of $\sum |a_n|$. For example, $\sum (-1)^n/n$ converges but $\sum 1/n$ diverges. If $\sum a_n$ converges but $\sum |a_n|$ diverges, then series $\sum a_n$ is said to converge conditionally.

Let (f_n) be a sequence of functions defined on some interval $[a, b]$. We can form a series $\sum f_n(x)$ for every $x \in [a, b]$. One says that the series $\sum f_n$ converges *pointwise* on $[a, b]$ if for the series $\sum f_n(x)$ converges for every point $x \in [a, b]$. That is, for each $x \in [a, b]$,

$$\left| f(x) - \sum_{k=1}^n f_k(x) \right| \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

If

$$\max_{x \in [a, b]} \left| f(x) - \sum_{k=1}^n f_k(x) \right| \rightarrow 0 \quad \text{as } n \rightarrow \infty,$$

then, the series $\sum f_n$ is said to converge *uniformly* on $[a, b]$.

Theorem 7.5 (Weierstrass M-test). *Let (f_n) be a sequence of functions on $[a, b]$ and let (M_n) be a sequence of positive constants satisfying:*

- (a) $|f_n(x)| \leq M_n$ for all $x \in [a, b]$ and all n ,
- (b) the series $\sum M_n$ converges.

The the series $\sum f_n$ converges absolutely and uniformly on $[a, b]$.

Theorem 7.6. *Let (f_n) be a sequence of functions on $[a, b]$ converging pointwise to the function f , i.e., $f(x) = \sum_{n=1}^{\infty} f_n(x)$.*

- (a) *If the f_n 's are continuous and the series $\sum f_n$ converges uniformly on $[a, b]$, then its sum f is a continuous function on $[a, b]$, and*

$$\sum_{n=1}^{\infty} \int_a^b f_n(x) dx = \int_a^b \sum_{n=1}^{\infty} f_n(x) dx = \int_a^b f(x) dx.$$

- (b) *If the f_n 's are of class C^1 and the series $\sum f'_n$ of derivatives converges uniformly on $[a, b]$, then its sum f is of class C^1 and*

$$f'(x) = \left(\sum_{n=1}^{\infty} f_n(x) \right)' = \sum_{n=1}^{\infty} f'_n(x).$$